

Lab Exercise 5: Taxicab Numbers

Objective

To write a Java program that identifies all **Taxicab numbers** up to a given limit **N** — i.e., numbers that can be expressed as the sum of two cubes in two or more distinct ways.

Aim

To develop a Java program that computes and displays all integers $\leq N$ that can be represented as the sum of two cubes in at least two different ways.

Theory

A **Taxicab number** is a positive integer that can be expressed as the sum of two positive cubes in two distinct ways.

The concept originates from a famous anecdote involving mathematicians **G. H. Hardy** and **S. Ramanujan**, where Ramanujan identified 1729 as the smallest such number.

Example:

$$1729 = 1^3 + 12^3 = 9^3 + 10^3$$

General Idea:

- For each pair of integers (a, b), compute $a^3 + b^3$.
 - Use a data structure to record which pairs lead to the same cube sum.
 - If any sum appears **with at least two distinct pairs**, that sum is a **Taxicab number**.
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Algorithm / Procedure

1. Input an integer **N** from the user.
2. Initialize a `HashMap<Integer, List<String>>` to map the sum of cubes to the pairs producing it.
3. For each pair (a, b) where $1 \leq a \leq b$, compute:

$$sum = a^3 + b^3$$

4. If $sum \leq N$, store the pair (a,b) in the map.
 5. After processing all pairs, print all sums (keys) that have **two or more pairs** associated with them.
 6. Display results in the format:
 7. number = (a1,b1) (a2,b2)
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Program Code

```
import java.util.*;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter N: ");
        int N = sc.nextInt();

        // TODO: Your code here
        // 1. Use nested loops for a, b
        // 2. Store sums of cubes in a HashMap
        // 3. Print numbers with at least 2 distinct pairs
    }
}
```

Sample Output

Input:

Enter N: 20000

Output:

1729 = (1,12) (9,10)

4104 = (2,16) (9,15)

13832 = (2,24) (18,20)

...

Evaluation Criteria (Total Marks: 10)

Test	Description	Marks
Program compiles and runs correctly	No syntax or runtime errors	2
Correct logic for computing cube sums	Proper implementation of nested loops	2
Correct use of HashMap to store pairs	Efficient data handling	2

Test	Description	Marks
Correct identification of multiple cube pairs	Matches expected Taxicab numbers	2
Output format and readability	Clear, properly formatted output	2
Total		10

Result

The program successfully finds and prints all **Taxicab numbers** up to the given limit **N**, displaying each number along with its distinct cube pairs.